

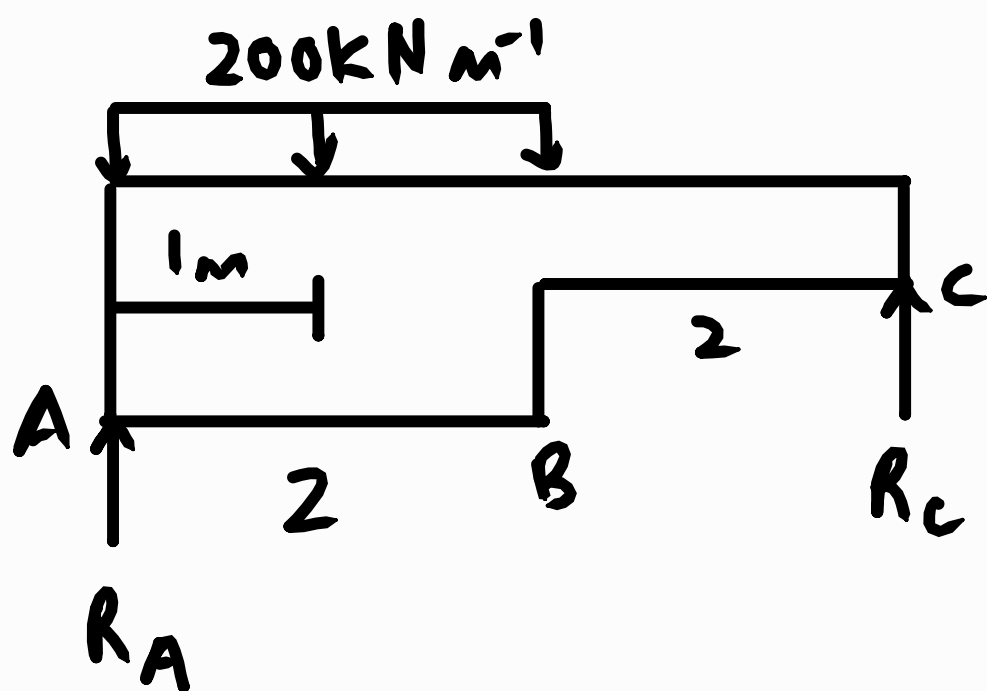
$$\begin{aligned}
 1) E &= 200 \text{ kN mm}^{-2} \\
 &= \frac{200 \times 10^3}{(1 \times 10^{-3})^2} \text{ N m}^{-2} \\
 &= 2 \times 10^{11} \text{ N m}^{-2}
 \end{aligned}$$

$$I_{AB} = 3 \times 10^6 (1 \times 10^{-3})^4 = 3 \times 10^{-6} \text{ m}^4$$

$$I_{BC} = 10^6 (1 \times 10^{-3})^4 = 1 \times 10^{-6} \text{ m}^4$$

$$\Delta = \delta U^*$$

$$\Delta = \int_0^{L_{AB}} \frac{M_{AB} m_{AB}}{EI_{AB}} dx + \int_0^{L_{BC}} \frac{M_{BC} m_{BC}}{EI_{BC}} dx$$



$$\sum M_A = 0$$

$$2 \times 2 \times 1 = R_C (4)$$

$$4R_C = 4 \text{ kN}$$

$$R_C = 1 \text{ kN}$$

$$\sum F_y = 0$$

$$R_A + R_C = 4 \text{ kN}$$

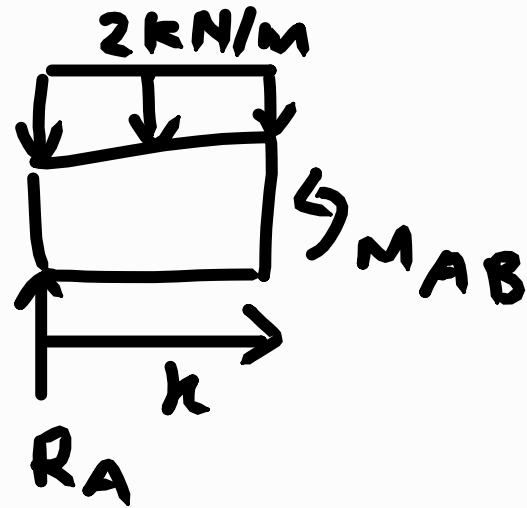
$$R_A = 4 - 1 = 3 \text{ kN}$$

1) For segment AB,

$$\sum M = 0$$

$$R_A x - 2x \left(\frac{x}{2} \right) = M_{AB}$$

$$M_{AB} = 3000x - 1000x^2$$

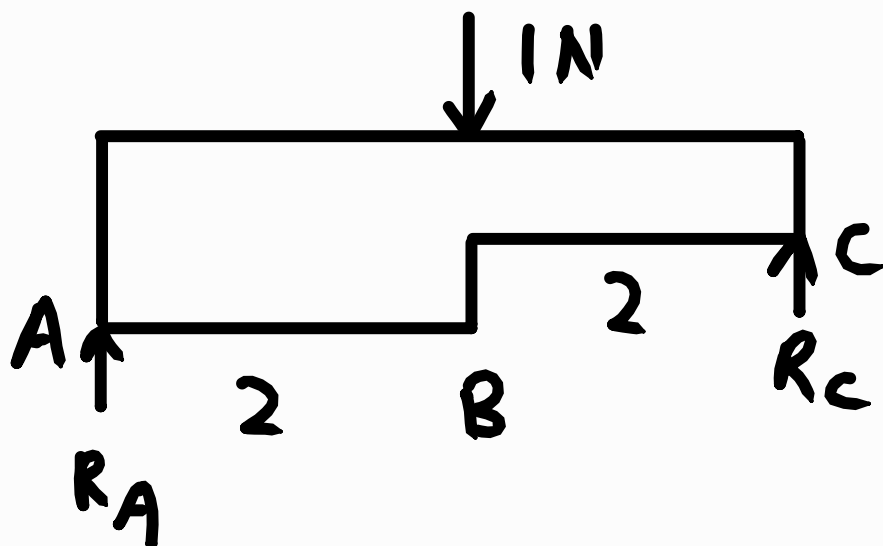
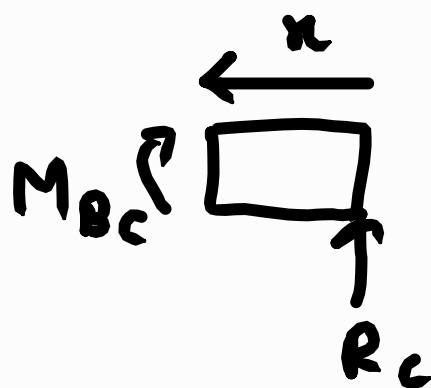


For segment BC,

$$\sum M = 0$$

$$R_c x = M_{BC}$$

$$M_{BC} = 1000x$$



$$\sum M_A = 0$$

$$4R_c = 2(1)$$

$$R_c = \frac{1}{2}$$

$$\sum F_y = 0$$

$$R_c + R_A = 1$$

$$R_A = 1 - \frac{1}{2} = \frac{1}{2}$$

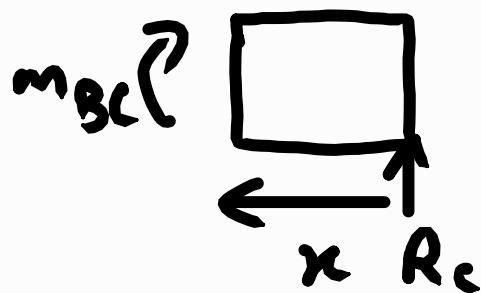
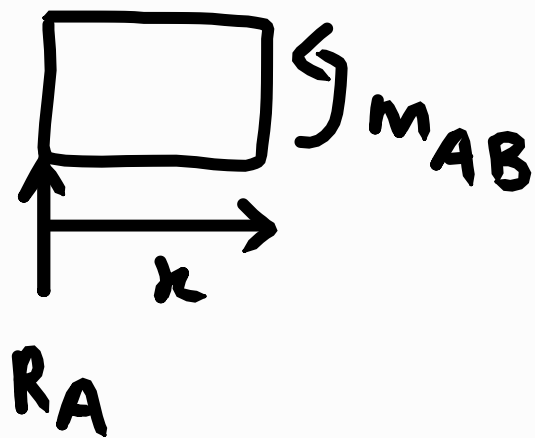
1) Segment AB:

$$\sum M = 0$$

$$M_{AB} = R_A x$$

$$m_{AB} = \frac{1}{2} x$$

Segment BC:



$$m_{BC} = R_C x$$

$$m_{BC} = \frac{1}{2} x$$

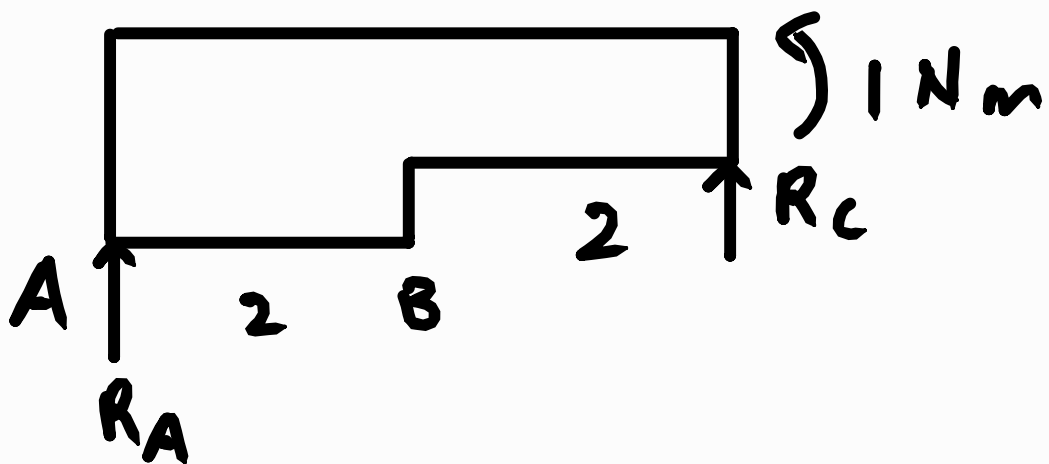
$$\therefore M_{AB} = 3000x - 1000x^2, M_{BC} = 1000x, m_{AB} = \frac{1}{2}x, m_{BC} = \frac{1}{2}x$$

$$\Delta = \int_0^{L_{AB}} \frac{M_{AB} m_{AB}}{EI_{AB}} dx + \int_0^{L_{BC}} \frac{M_{BC} m_{BC}}{EI_{BC}} dx$$

$$= \int_0^2 \frac{(3000x - 1000x^2)(\frac{1}{2}x)}{2 \times 10^{11} (3 \times 10^{-6})} dx + \int_0^2 \frac{(1000x)(\frac{1}{2}x)}{2 \times 10^{11} (1 \times 10^{-6})} dx$$

$$= 0.01 \text{ m downwards}$$

1)



$$\sum M_C = 0$$

$$4R_A = 1$$

$$R_A = \frac{1}{4} \text{ Nm}$$

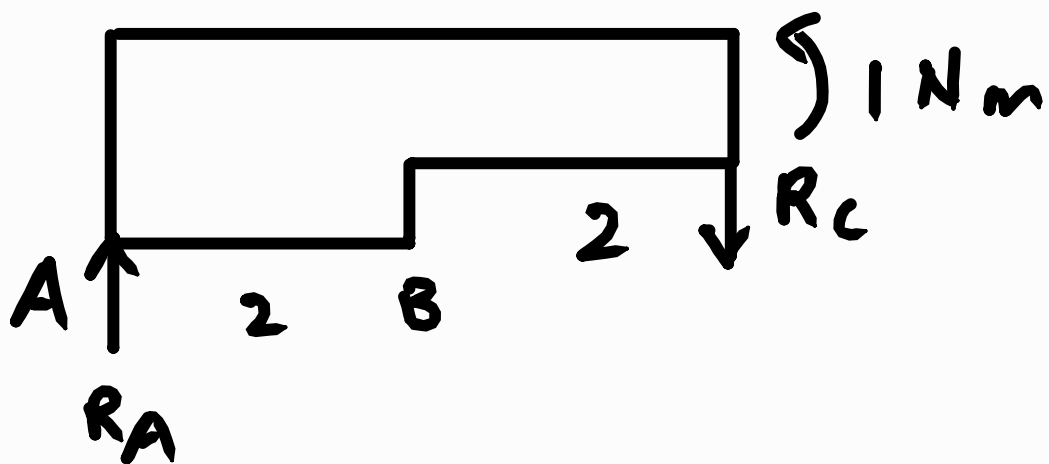
$$\sum F_y = 0$$

$$R_A + R_C = 0$$

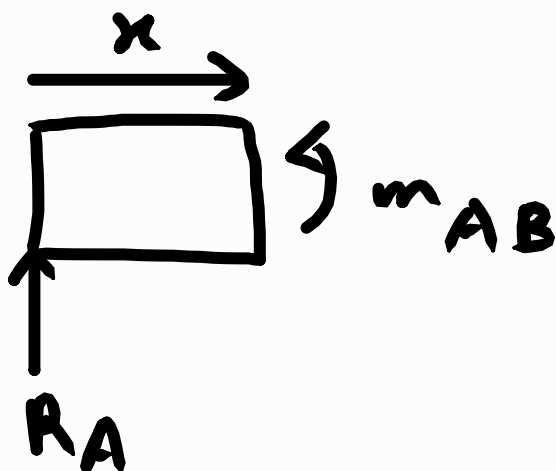
$$R_C = -R_A$$

$$= -\frac{1}{4} \text{ Nm}$$

$$\therefore R_C \downarrow$$



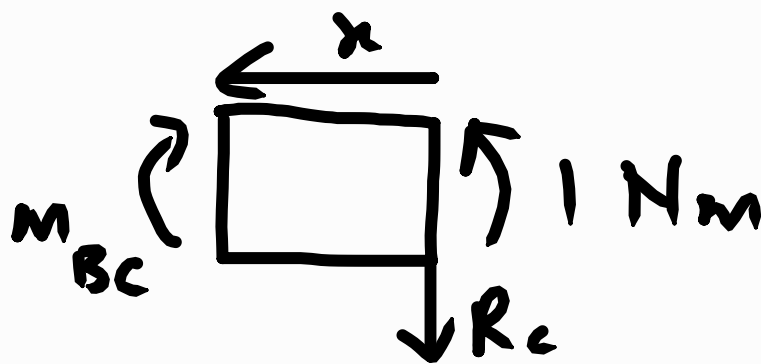
Segment AB:



$$m_{AB} = R_A$$

$$= \frac{1}{4}x$$

Segment BC:



$$m_{BC} + R_C x = 1$$

$$M_{BC} = 1 - \frac{1}{4}x$$

$$\therefore M_{AB} = 3000x - 1000x^2, M_{BC} = 1000x, m_{AB} = \frac{1}{4}x, m_{BC} = 1 - \frac{1}{4}x$$

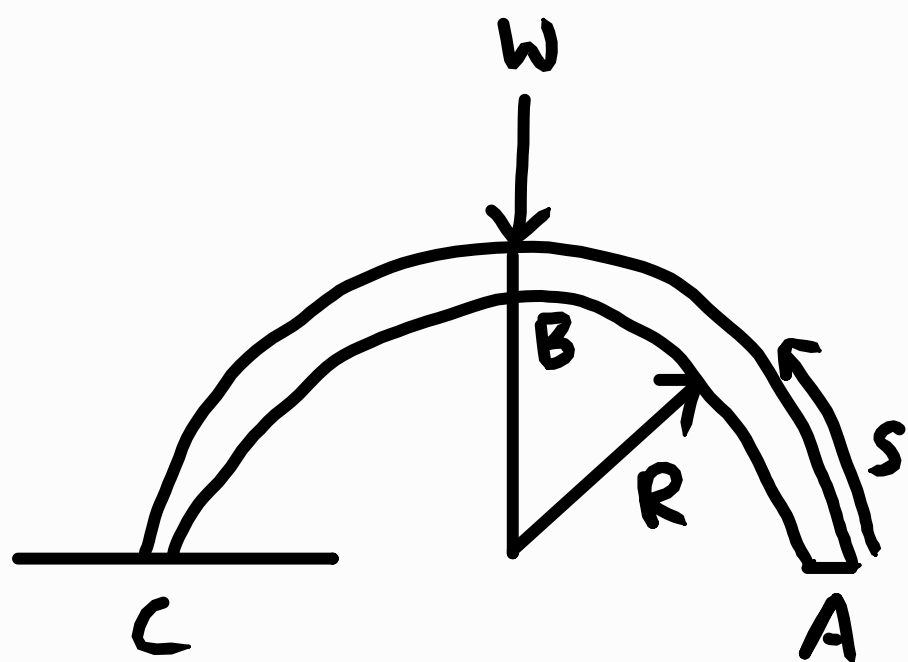
$$1) \theta_c = \int_0^{L_{AB}} \frac{M_{AB} m_{AB}}{EI_{AB}} dx + \int_0^{L_{BC}} \frac{M_{BC} m_{BC}}{EI_{BC}} dx$$

$$= \int_0^2 \frac{(3000x - 1000x^2)(\frac{1}{4}x)}{2 \times 10^{11} (3 \times 10^{-6})} dx + \int_0^2 \frac{(1000x)(1 - \frac{1}{4}x)}{2 \times 10^{11} (1 \times 10^{-6})} dx$$

$$= \frac{1}{120} \text{ rad}$$

$$\approx 0.477^\circ$$

2)

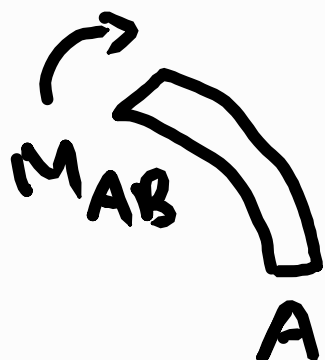


$$s = R\theta \quad ds = R d\theta$$

$$\Delta = \delta U^*$$

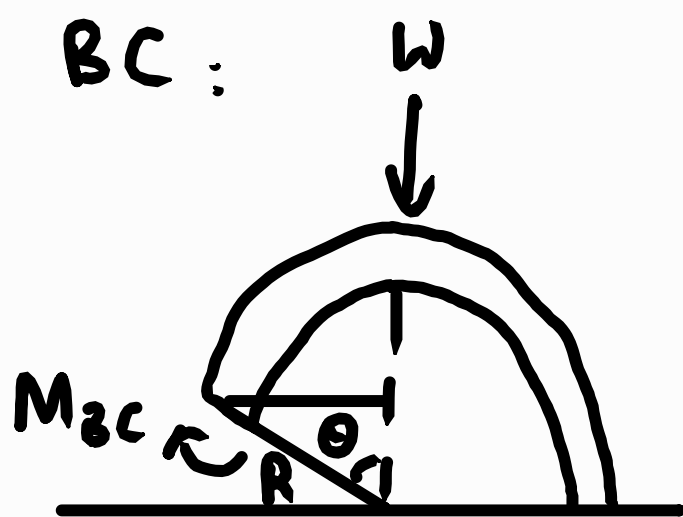
$$= \int_0^{\frac{\pi}{2}} \frac{M_{AB} m_{AB}}{EI} R d\theta + \int_0^{\frac{\pi}{2}} \frac{M_{AB} m_{AB}}{EI} R d\theta$$

Segment AB:



$$M_{AB} = 0$$

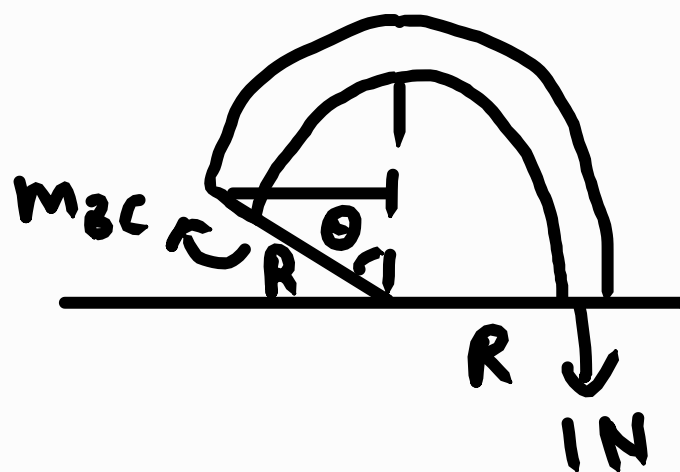
Segment BC:



$$M_{BC} = W(R \sin \theta)$$

$$= WR \sin \theta$$

Unit vertical load:

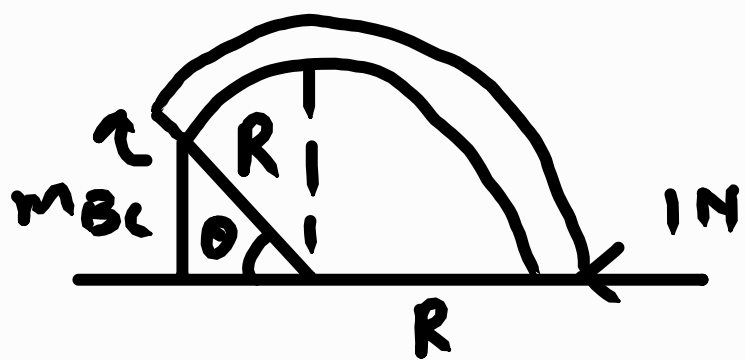


$$m_{BC} = 1(R + R \sin \theta)$$

$$= R + R \sin \theta$$

$$\begin{aligned}
 2) \Delta_v &= \int_0^{\frac{\pi}{2}} \frac{\cancel{M_{AB}}^0 \cancel{m_{AB}}}{EI} R d\theta + \int_0^{\frac{\pi}{2}} \frac{M_{BC} m_{BC}}{EI} R d\theta \\
 &= \int_0^{\frac{\pi}{2}} \frac{WR \sin \theta (R + R \sin \theta)}{EI} R d\theta \\
 &= \frac{WR^3}{EI} \int_0^{\frac{\pi}{2}} \sin \theta (1 + \sin \theta) d\theta \\
 &= \frac{WR^3}{EI} \left(1 + \frac{1}{4} \pi \right)
 \end{aligned}$$

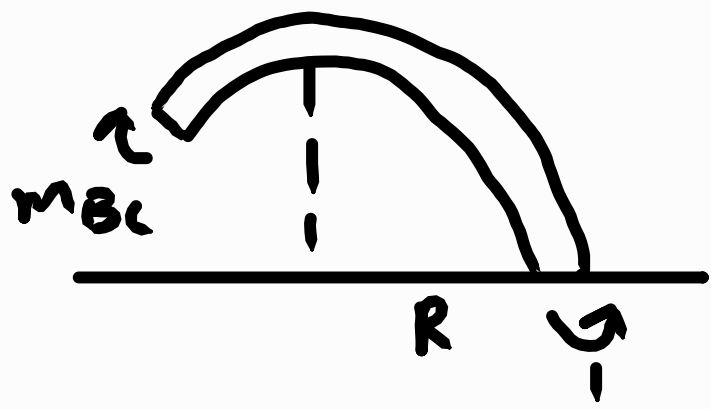
Unit horizontal load:



$$\begin{aligned}
 m_{BC} &= 1(R \cos \theta) \\
 &= R \cos \theta
 \end{aligned}$$

$$\begin{aligned}
 \Delta_h &= \int_0^{\frac{\pi}{2}} \frac{M_{BC} m_{BC}}{EI} R d\theta \\
 &= \int_0^{\frac{\pi}{2}} \frac{WR \sin \theta R \cos \theta}{EI} R d\theta \\
 &= \frac{WR^3}{EI} \int_0^{\frac{\pi}{2}} \sin \theta \cos \theta d\theta \\
 &= \frac{WR^3}{EI} \left(\frac{1}{2} \right) = \frac{WR^3}{2EI}
 \end{aligned}$$

2) Unit moment:



$$m_{Bc} = 1$$

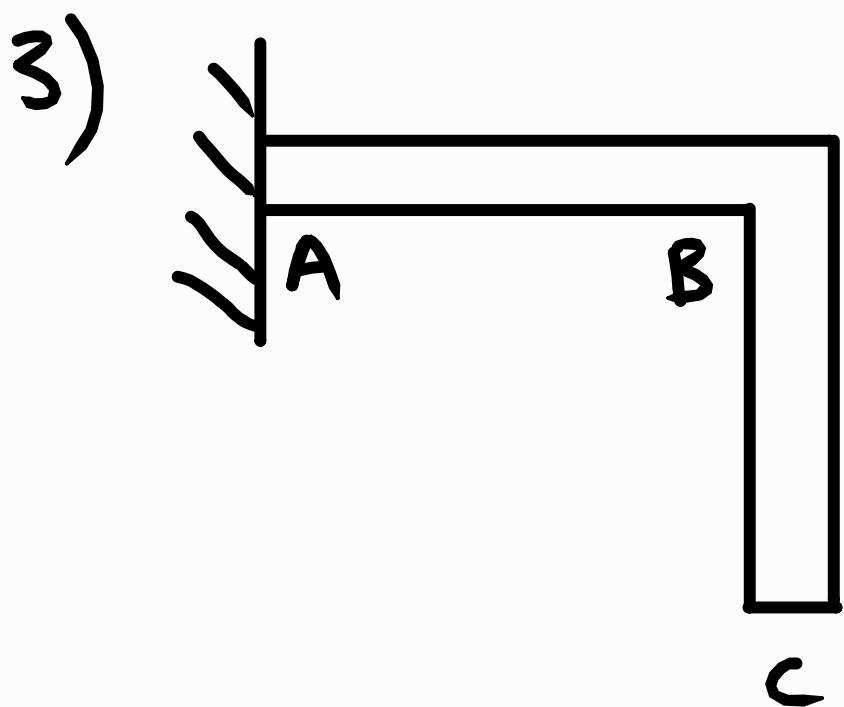
$$\theta_A = \int_0^{\frac{\pi}{2}} \frac{WR \sin \theta (1)}{EI} R d\theta$$

$$= \frac{WR^2}{EI} \int_0^{\frac{\pi}{2}} \sin \theta d\theta$$

$$= \frac{WR^2}{EI} \left[-\cos \theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{WR^2}{EI} \left[-0 - (-1) \right]$$

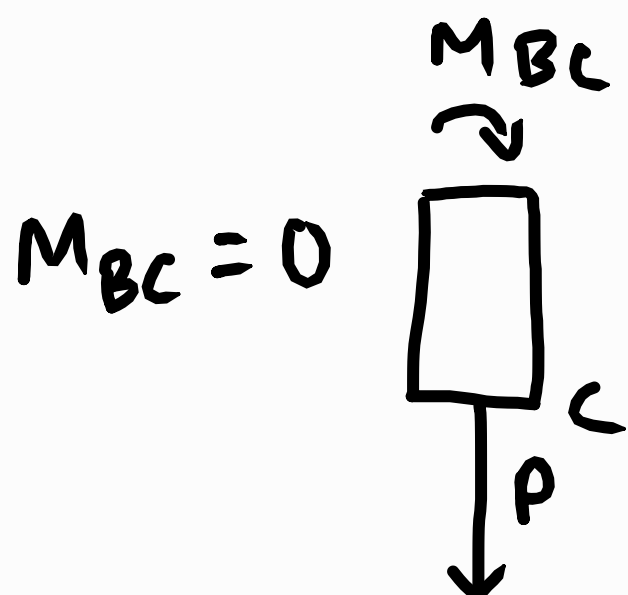
$$= \frac{WR^2}{EI}$$



$$\Delta = \int U^*$$

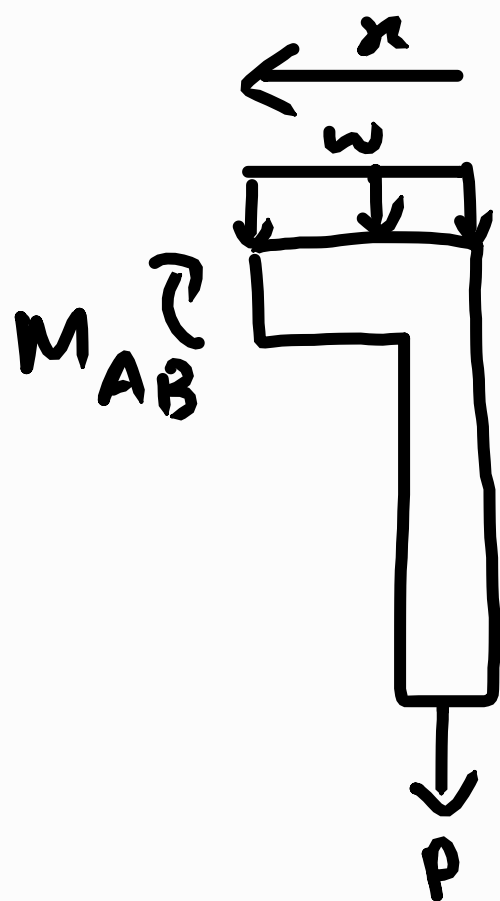
$$= \int_0^L \frac{M_{BC} m_{BC}}{EI} dx + \int_0^L \frac{M_{AB} + m_{AB}}{EI} dx$$

Segment BC:

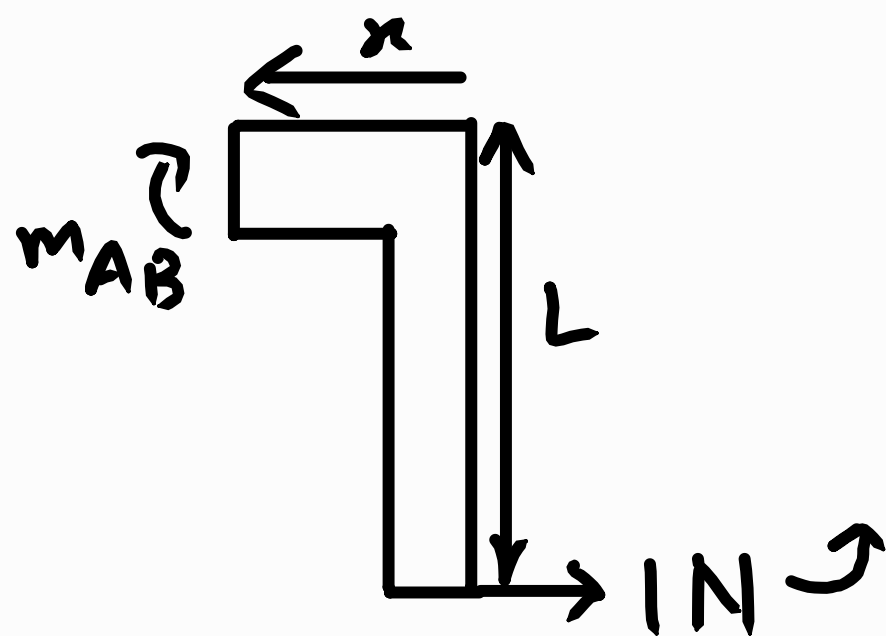


$$\begin{aligned} \therefore \Delta &= \int_0^L \frac{\overset{0}{M_{BC}} m_{BC}}{EI} dx + \int_0^L \frac{M_{AB} + m_{AB}}{EI} dx \\ &= \int_0^L \frac{M_{AB} m_{AB}}{EI} dx \end{aligned}$$

3) Segment AB:



Unit horizontal load



$$\therefore M_{AB} = L$$

$$M_{AB} + Px + wx\left(\frac{x}{2}\right) = 0$$

$$M_{AB} = -Px - \frac{wx^2}{2}$$

$$\Delta = \int_0^L \frac{(-Px - \frac{wx^2}{2})(L)}{EI} dx$$

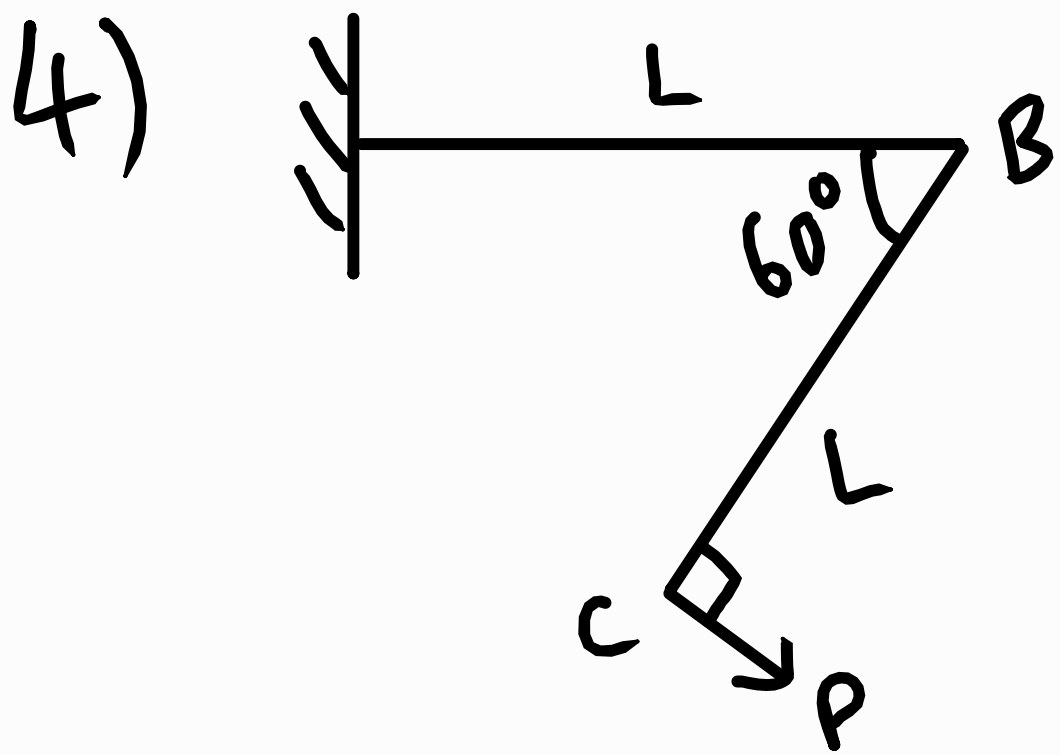
$$= -\frac{L}{EI} \int_0^L Px + \frac{wx^2}{2} dx$$

$$= -\frac{L}{EI} \left[\frac{Px^2}{2} + \frac{wx^3}{6} \right]_0^L$$

$$= -\frac{L}{EI} \left[\frac{PL^2}{2} + \frac{wL^3}{6} \right]$$

$$= -\frac{L^3}{2EI} \left[P + \frac{wL}{2} \right]$$

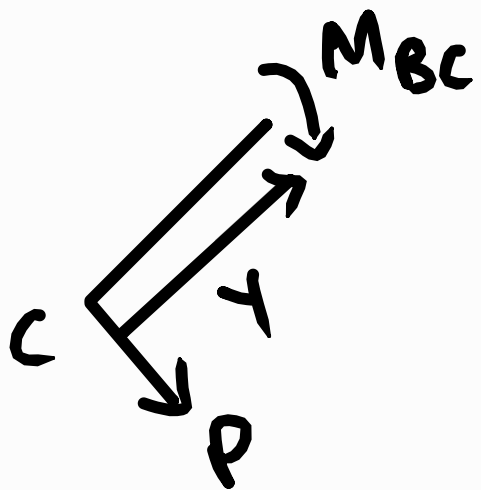
$\therefore$ the horizontal deflection is $\frac{L^3}{2EI} \left[P + \frac{wL}{2} \right]$ to the left.



$$\Delta = \delta U^*$$

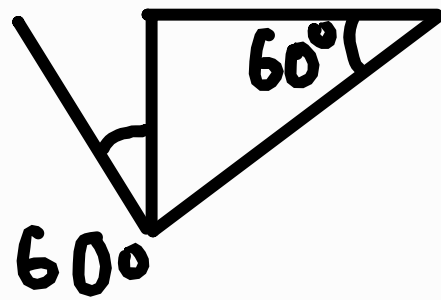
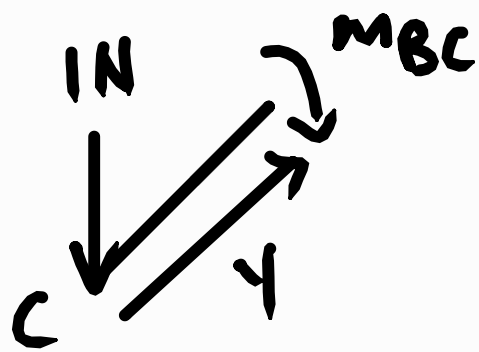
$$= \int_0^L \frac{M_{AB} m_{AB}}{EI} dx + \int_0^L \frac{M_{BC} m_{BC}}{EI} dy$$

Segment BC:



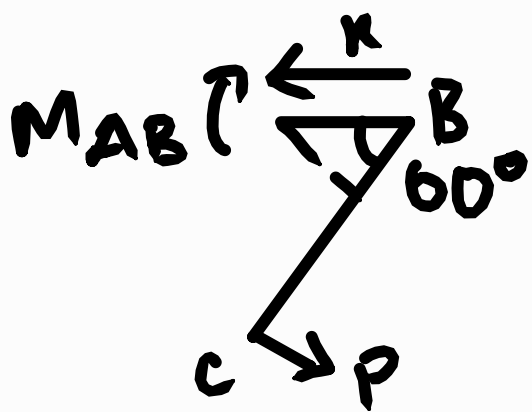
$$M_{BC} = Py$$

Unit vertical load



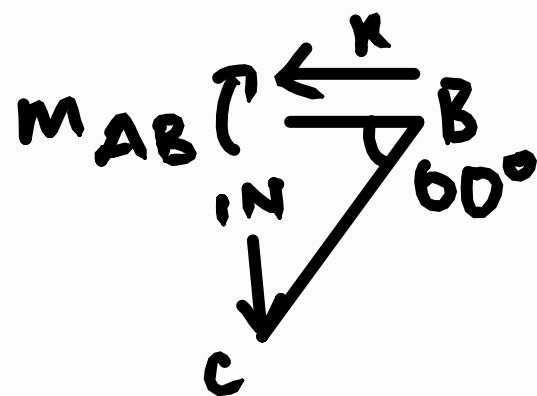
$$m_{BC} = y(1 \cos 60^\circ) = \frac{y}{2}$$

Segment AB:



$$M_{AB} = P(L - x \cos 60^\circ) = P(L - \frac{x}{2})$$

Unit vertical load



$$m_{AB} = 1(L \cos 60^\circ - x) = \frac{L}{2} - x$$

$$\begin{aligned}
4) \Delta &= \int_0^L \frac{M_{AB} m_{AB}}{EI} dx + \int_0^L \frac{M_{BC} m_{BC}}{EI} dy \\
&= \int_0^L \frac{p(L - \frac{x}{2})(\frac{L}{2} - x)}{EI} dx + \int_0^L \frac{py(\frac{y}{2})}{EI} dy \\
&= \frac{p}{EI} \left[\int_0^L (L - \frac{x}{2})(\frac{L}{2} - x) dx + \int_0^L \frac{y^2}{2} dy \right] \\
&= \frac{p}{EI} \left[\int_0^L \left(\frac{L^2}{2} - Lx - \frac{Lx}{4} + \frac{x^2}{2} \right) dx + \left[\frac{y^3}{6} \right]_0^L \right] \\
&= \frac{p}{EI} \left[\left[\frac{L^2}{2}x - \frac{5Lx^2}{8} + \frac{x^3}{6} \right]_0^L + \frac{L^3}{6} \right] \\
&= \frac{p}{EI} \left[\left(\frac{L^3}{2} - \frac{5L^3}{8} + \frac{L^3}{6} \right) + \frac{L^3}{6} \right] \\
&= \frac{p}{EI} \left(\frac{5}{24} L^3 \right) \\
&= \frac{5pL^3}{24EI} \text{ (downwards)}
\end{aligned}$$